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Surface Areas and Volumes

Shapes	Curved Surface area	Total Surface are	Volume
Cube	$4l^2$	$6l^2$	l^3
Cuboid	$2(lb+hl)$	$2(lb+bh+hl)$	lbh
Cylinder	$2\pi rh$	$2\pi rh+\pi r^2$	$\frac{1}{3}\pi r^2 h$
Cone	πrl	$\pi r(l+r)$	$\frac{1}{3}\pi r^2 h$
Hemisphere	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$
Sphere	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$

Introduction:-

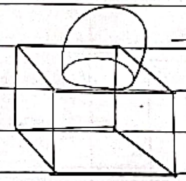
- Combination of solids:

Tent-

→ Cone
+ = Tent
→ cylinder

Ice-creame cone-

→ Hemisphere
+ = Ice-cream cone
→ Cone



→ Hemisphere
+
→ Cube

Ex-12.1

1-Sol: Let side of the cube = x cm

Volume of the cube = $64\ cm^3$

$\therefore\ x^3 = 64$

$x = 4$ cm

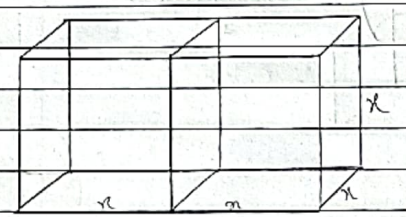
Thus, the side of the cube = 4 cm

When two cubes are joined end to end, a cuboid is formed.

Length of the cuboid = $2x = 8$ cm

Width of the cuboid = $x = 4$ cm

Height of the cuboid = $x = 4$ cm



$\therefore$ Surface area of the cuboid

$= 2(lb + bh + hl)$

$= 2(8\times4 + 4\times4 + 4\times8)$

$= 2(32 + 16 + 32)$

$= 2(80) = 160$

Hence, the surface area of the cuboid is $160\ cm^2$.

2-Sol: Diameter of the hemisphere = Diameter of the cylinder = 14 cm

$\therefore$ Radius of the hemisphere $= \frac{14}{2} = 7$ cm

Total height of the vessel = 13 cm

$\therefore$ Height of the cylinder $= 13 - 7 = 6$ cm

Inner surface area of the vessel = Inner surface area of the hemisphere + Inner surface area of the cylinder $= 2\pi r^2 + 2\pi rh$

$= 2\pi r(r+h)$

$= 2\times\frac{22}{7}\times7(7+6)$

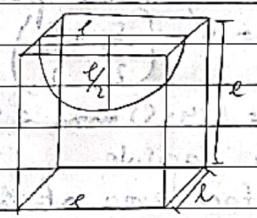
$= 44\times13 = 572\ cm^2$

Hence, inner surface area of the vessel is $572\ cm^2$.

5-Sol: Diameter of the hemisphere = Edge of the cube

i.e. $2r = l$

$r = \frac{l}{2}$



Surface area of the remaining solid

= Total surface area of the cube

+ Inner surface area of the hemisphere

− Area of base of hemisphere.

$= 6(l)^2 + 2\pi r^2 - \pi r^2$

$\Rightarrow 6(l)^2 + \pi r^2$

$\Rightarrow 6(l)^2 + \pi\left(\frac{l}{2}\right)^2$

$\Rightarrow \frac{l^2}{4}(24 + \pi)$

$\Rightarrow \frac{l^2}{4}(\pi + 24)$ sq. units

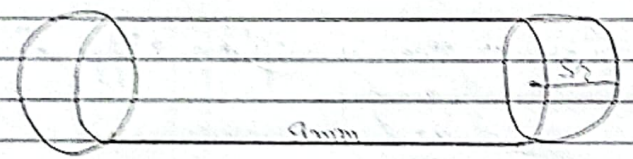
6-Solu Diameter of the capsule

= Diameter of the hemisphere

= Diameter of cylinder = 5mm

$2r = 5$ mm

$r = \frac{5}{2}$ mm



Length of entire capsule = 14mm

∴ Length of the cylindrical part

$= 14 - \left(\frac{5}{2} + \frac{5}{2}\right)$

$= (14 - 5)$ mm $= 9$ mm

∴ Surface area of the capsule

= Surface area of the cylinder

+ 2 (Surface area of one hemisphere)

$= 2\pi rh + 2(\pi r^2)$

$= 2\pi r(h + 2r)$

$= 2 \times \frac{22}{7} \times \frac{5}{2}\left(9 + 2\left(\frac{5}{2}\right)\right)$

$= \frac{22}{7} \times 5 \times 14 = 220$ mm²

Hence, the surface area of the capsule is 220 mm².

7-Solu Diameter of the cylinder

= Diameter of the cone = 1.4cm

∴ Radius of the cylinder = Radius of the cone

= 0.7cm

Hieght of the cylinder = 2.4 cm

Hieght of cone = Hieght of cylinder = 2.4cm

Since, $l^2 = r^2 + h^2$

$l^2 = (0.7)^2 + (2.4)^2 = 0.49 + 5.76$

$l^2 = 6.25$

$l = 2.5$ cm

∴ Total surface area of the remaining solid

= Curved surface area of cylinder

+ Area of the base of cylinder +

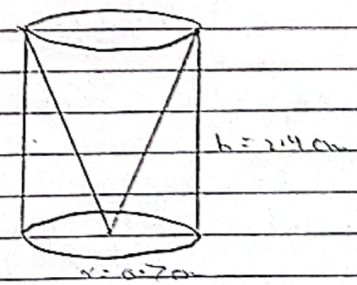
Surface area of the cone

$= 2\pi rh + \pi r^2 + \pi r l = \pi r(2h + r + l)$

$= \frac{22}{7} \times 0.7(2(2.4) + 0.7 + 2.5)$

$= 22 \times 0.1(4.8 + 3.2) = \frac{22}{10} \times 8 = 17.6$ cm²

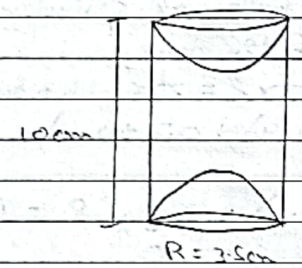
Hence, the total surface area of the remaining solid is 18 cm².



9-Solu Hight of the cylinder = 10 cm
Radius of the cylinder = Radius of the hemisphere = 3.5 cm

∴ Surface area of the remaining article
= Curved surface area of cylinder + 2 (Curved surface area of the hemisphere)
$= 2\pi rh + 2(2\pi r^2) = 2\pi r(h + 2r)$
$= 2 \times \frac{22}{7} \times 3.5 (10 + 2(3.5)) = \frac{44}{7} \times 0.5 (12)$
$= 374 \text{ cm}^2$

Hence, the total surface area of the article is 374 cm^2.



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Ex-12.2

1-Solu Here radius of the cone = Radius of the hemisphere = 1cm

Thus, $r = 1$ cm

Height of the cone (h) = 1cm
∴ Volume of the Solid = Volume of the cone + Volume of hemisphere

$= \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3 = \frac{1}{3}\pi r^2 (h + 2r)$
$= \frac{1}{3} \times \frac{22}{7} \times (1)^2 (1 + 2) = \frac{22}{7} \times 3 = \pi$

Hence, the volume of the solid is 3.14 cm^3

2-Solu Radius of the cone = Radius of the cylinder (r) $= \frac{3}{2}$ cm

Thus, $r = 1.5$ cm
Height of each cone = 2 cm (h)
Height of the cylinder = 12 − (2+2) = 8 cm (H)
∴ Volume of air in the model = Volume of the cylinder + 2 (Volume of the cone)
$= \pi r^2 H + 2\left(\frac{1}{3}\pi r^2 h\right)$
$= \pi r^2 \left(H + \frac{2}{3}h\right)$
$= \frac{22}{7} \times 1.5 \times 1.5 \left(8 + \frac{2}{3}(2)\right)$
$= \frac{22}{7} \times \frac{3}{2} \times \frac{3}{2} \times \frac{28}{3} = 66$

Hence, the volume of air in the model is 66 cm^3

4-Sol. Here, length of cuboid (l) = 15 cm
width of cuboid (b) = 10 cm
hieght of cuboid (h) = 3.5 cm
Also radius of conical cavity (R) = 0.5 cm
and hieght of conical cavity (H) = 1.4 cm
∴ Volume of wood in the entire stand =
Volume of cuboid − 4 (volume of cone)

$= lbh - 4\left(\frac{1}{3}\pi R^2 H\right)$

$= 15 \times 10 \times 3.5 - 4\left(\frac{1}{3} \times \frac{22}{7} \times 0.5 \times 0.5 \times 1.4\right)$

$= 15 \times \cancel{10} \times \frac{35}{\cancel{10}} - 4\left(\frac{1}{3} \times \frac{\cancel{22}^{11}}{7} \times \frac{\cancel{5}^{1}}{\cancel{10}_{2}} \times \frac{\cancel{5}^{1}}{\cancel{10}_{2}} \times \frac{\cancel{14}^{7}}{10}\right)$

$= 15 \times 35 - 4\left(\frac{1}{3} \times 11 \times \frac{1}{10}\right)$

$= 15 \times 35 - 4\left(\frac{11}{30}\right)$

$= 15 \times 35 - \frac{44}{30} = 525 - \frac{\cancel{44}^{22}}{\cancel{30}_{15}} = 525 - \frac{22}{15}$

$= 525 - 1.466 = 523.53 \text{ cm}^3$

Hence the volume of wood in the entire stand is 523.53 cm³.

6-Sol. Radius of Lower Cylinder (R) $= \frac{1}{2}(24) = 12$ cm

Height of Lower cylinder (H) = 220 cm
Radius of upper cylinder (r) = 8 cm
Height of upper cylinder (h) = 60 cm

∴ Volume of the pole = Volume of lower cylinder + Volume of upper cylinder

$= \pi R^2 H + \pi r^2 h$

$= 3.14(12 \times 12 \times 220 + 8 \times 8 \times 60)$

$= 3.14(3168 + 3840)$

$= 3.14(7008) = 111532.8$

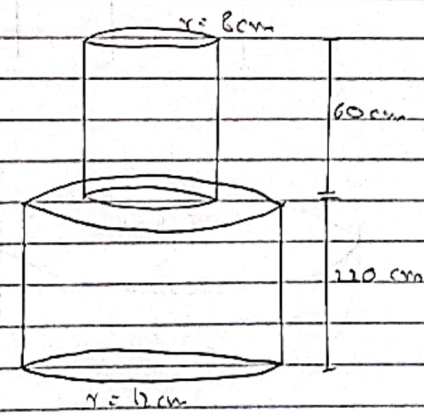
Mass of 1 cm³ = 8 g

∴ Mass of 111532.8 cm³ $= 8 \times 111532.8$

$= \frac{892262.4}{1000}$

$= 892.2624$ kg

Hence, the mass of the pole is 892.2624 kg.



8-Solu Radius of the neck (r) = $\frac{2}{2}$ = 1 cm

Height of the cylindrical portion (h) = 8 cm

Radius of spherical portion (R) = $\frac{8.5}{2}$ = 4.25 cm

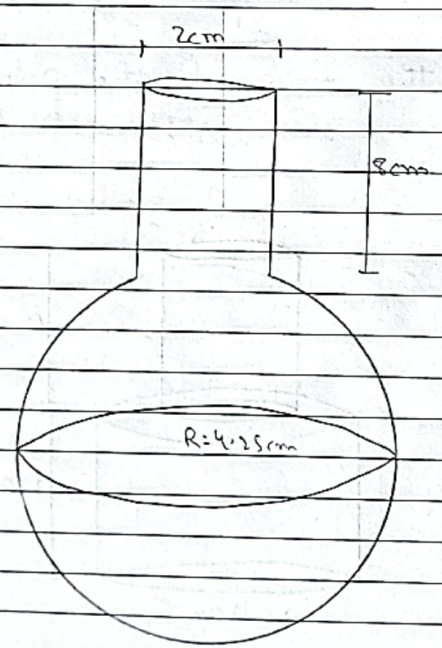
∴ Volume of water in the vessel = volume of cylinder + volume of sphere.

$= \frac{4}{3}\pi R^3 + \pi r^2 h$

$= \frac{4}{3} \times 3.14 \times 4.25 \times 4.25 \times 4.25 + 3.14 \times 1 \times 1 \times 8$

$= 321.39 + 25.12 = 346.51 \text{ cm}^3$

Hence, the child is not correct.



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